

Are Algorithmic Maps Partisan?

Situating a BISECT Benchmark Within ReCom Ensembles

Two Implementations and a Complete 2020 Election Model

G.2 — Partisan Outcome Distributions

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Abstract

We report real tract-level partisan-seat distributions for Rhode Island, Iowa, and North Carolina using the 2020 presidential estimates and two ReCom implementations. Rhode Island has no seat variation: the two-seat benchmark and all sampled plans elect two Democratic-majority districts under this model. Iowa’s one-Democratic-seat benchmark falls at the 34th–37th percentile. North Carolina’s five-seat benchmark falls at the 17th–21st percentile, but ESS is below 100 for both implementations and the result is not a converged tail finding. The locally available 2016 estimates fail current-tract coverage and are excluded from the primary results. These findings do not support the former six-state “near median” generalization.

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1. Election Model And Coverage

The primary metric sums tract-level Democratic and Republican presidential estimates within each district and counts Democratic-plurality districts. The 2020 file matched every Rhode Island, Iowa, and North Carolina benchmark tract. This is a descriptive seats model, not a legal fairness test [Stephanopoulos and McGhee, 2015, Rodden, 2019]. The 2016 file missed 12 of 250 Rhode Island tracts, 129 of 896 Iowa tracts, and 942 of 2,672 North Carolina tracts because it does not share the 2020 tract universe. No crosswalk was available, so 2016 is excluded from the primary percentiles rather than treating unmatched units as zero votes.

2. 2020 Presidential Seat Results

Table 1: Benchmark Democratic-seat position under the 2020 presidential tract model.

State	Tool	Benchmark	Mean	Percentile	$\hat{R}$	ESS
RI	Rust	2	2.000	50.0	1.000	6000
RI	GerryChain	2	2.000	50.0	1.000	6000
IA	Rust	1	1.266	36.7	1.002	486
IA	GerryChain	1	1.323	33.9	1.006	1072
NC	Rust	5	5.944	17.1	1.028	90
NC	GerryChain	5	5.811	21.2	1.036	98

Rhode Island is degenerate for this metric: every sampled plan has two Democratic seats, so the mid-rank percentile is 50% but carries no evidence of partisan neutrality. Iowa is below the ensemble mean but not in an extreme tail. North Carolina is lower than most sampled plans, but both ESS values fail the preregistered 100-sample threshold; the 17–21% estimates remain non-converged descriptive results.

3. Interpretation

The state results are heterogeneous and election-specific. They do not show that the benchmark systematically favors or disfavors a party. They also do not support the earlier claim that the algorithm is near the partisan median in five of six states. Cross-tool seat means are similar here, while the underlying cut distributions differ materially.

The proportionality corridor is not reported because the experiment did not preregister a statewide proportional-entitlement rule. A seat percentile under one presidential election is a diagnostic, not a fairness standard.

4. Claim Boundary

The package is tract-level, covers three states and one complete election, and does not establish intent, causation, partisan fairness, or legal validity. Additional elections require a valid crosswalk to the same unit universe.

References

Jonathan A. Rodden. Why cities lose: The deep roots of the urban-rural political divide. 2019.

Nicholas O. Stephanopoulos and Eric M. McGhee. Partisan gerrymandering and the efficiency gap. *University of Chicago Law Review*, 82:831–900, 2015.